

Syed Waleed Hyder

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Eligible for F-1 OPT/STEM OPT work authorization; no immediate sponsorship required

EXPERIENCE

- ID Privacy AI** Remote
 - Graduate Capstone Project — AI Agent Engineer* Sep 2026 – Present
 - LLM Agent Engineering & Optimization:** Engineered and iteratively optimized the prompt architecture, conversation workflows, guardrails, and tool-calling behavior of production AI service agents for automotive dealerships, increasing service appointment booking completion from 40% to 80% of production calls with booking intent and reducing hallucinations and incorrect tool usage from 30% to 5% of calls.
 - LLM Evaluation & Regression Testing:** Built and maintained a golden dataset of 100 multi-turn conversations covering service workflows, edge cases, state transitions, and tool-calling behavior; ran regression evaluations across agent changes to validate improvements and prevent degradation of existing capabilities.
- T3 - Wichita State University** Wichita, KS, US
 - Graduate Assistant - Machine Learning* Jul 2025 – Present
 - Vizling — Zero-Shot Image Segmentation:** Developed and evaluated a VLM-based zero-shot image annotation tool. Used Qwen3.5 for comprehensive scene understanding and structured annotation parsing, Google OWL-ViT for text-driven object detection, and Google PaliGemma for multi-object segmentation, iteratively validating outputs for accuracy before hand-off.
 - Vizling — Fine-Tuning Vision-Language Model:** Fine-tuned Qwen3.5-4B using LoRA on a custom image-to-JSON instruction dataset, collapsing the two-stage scene understanding and structured parsing pipeline into a single, end-to-end process; verified JSON schema conformance and output correctness across test cases.
- Rapidev AI** Remote
 - Machine Learning Engineer* Sep 2023 – Dec 2024
 - Agentic RAG System for Finance Compliance Intelligence:** Built and tested an AI-based compliance assistant using Agentic Retrieval-Augmented Generation (RAG) to identify and summarize finance regulations for new financial products. Developed a multi-step reasoning agent for semantic document search, compliance categorization, and citation-based summaries, validating outputs against source documents for accuracy.
 - Few-Shot User Intent Classification:** Built, evaluated, and deployed a multi-label intent classifier for a debt collection product using few-shot learning (DSPy + GPT-4, F1 = 0.85; RoBERTa + SetFit, F1 = 0.87), using cross-validation to benchmark and select the best-performing approach before shipping.
- Retrocausal** Redmond, Washington (Remote)
 - Research Engineer* Jul 2022 – Sep 2023
 - Retroactivity — Video Understanding Module:** Worked on supervised & unsupervised fine-grained human activity segmentation using both frames and skeletons (2D/3D pose). Implemented and tested an algorithm for unsupervised action segmentation by detecting action boundaries, which was validated and deployed on the iPad version of Retroactivity.
- OMNO AI** Lahore, Pakistan
 - Machine Learning Engineer* Sep 2021 – Jul 2022
 - SportsEye: AI-Powered Football Analytics Platform:** Utilized broadcast streams for player and ball tracking, eliminating extra hardware needs. Implemented Player & Ball Detection using Single Stage Detectors, player tracking & re-identification with DeepSORT. Developed Broadcast view classification with custom CNNs, Field Line Masking using Pix2Pix GANs, Camera Pose Estimation via classical Homography, and Timer Localization using EAST detector with OCR methods – validating each module's output before integration into the full pipeline.
 - Adlytic: Audience Analytics Platform for Retail Stores:** Developed and tested a pipeline for football counting, age and gender classification, area-wise heatmap generation, and dwell time analysis. Created and verified APIs for KPI dashboard integration, confirming data mapped correctly end-to-end. Deployed the system using Docker in over 50 retail stores, achieving above 90% accuracy using YOLOv5, DeepSORT, and BYTETrack.

- **Systems Limited** Karachi, Pakistan
Machine Learning Consultant *Aug 2020 – Aug 2021*
 - **Regeneron Pharmaceuticals: Data Analysis of Patient Wearable Devices:** Conducted gait analysis using ML algorithms on time-series data from Moticon sensors. Classified human activity based on accelerometer data from the Actigraph device using ML techniques, cross-checking model predictions against ground truth for quality control. Tools: PySpark, AWS Redshift
- **SiPEO, Technical University of Munich** Munich, Germany
Research Intern *Summer 2019*
 - **Slum Mapping in Satellite Imagery Using Deep Learning:** Collected and filtered image data on slums in Karachi and Islamabad. Utilized Fully Convolutional Networks (FCN) for segmenting slums from non-slums. Addressed class imbalance using specialized loss functions. Implemented transfer learning and adversarial domain adaptation to align embeddings of Karachi and Mumbai slum datasets.

SKILLS

- **LLM / Agent Engineering:** LLM Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering & Architecture, Agent Evaluation, Regression Testing, Tool Calling, Structured Outputs
- **ML / Vision:** PyTorch, TensorFlow, OpenCV, Object Detection, Semantic Segmentation, Transformer Models
- **Engineering:** Python, Docker, APIs, AWS, GPU Computing, Databases

PUBLICATION

- **Action Segmentation Using 2D Skeleton Heatmaps (ICRA 2024):** Developed a novel 2D skeleton-based action segmentation method using 2D skeleton heatmaps and Temporal Convolutional Networks, achieving comparable or superior results and robustness in fine-grained human activity recognition. This work is the first to utilize 2D skeleton heatmap inputs and explore 2D skeleton+RGB fusion for action segmentation.

EDUCATION

- **Wichita State University (WSU)** Wichita, Kansas
Master of Science in Computer Science; GPA: 4.0/4.0 *Jan 2025 – Dec 2026*
- **National University of Sciences and Technology (NUST)** Islamabad, Pakistan
Bachelor of Engineering in Software Engineering; GPA: 3.67/4.0 *Sep 2016 – May 2020*